Battletoads — Game Testing Study

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Game Testing Report Game: Battletoads July 23, 2022 Airron Kennedy Angelo Bowins Valerie Rubio Instructor Robert Kennedy

Introduction Description: Battletoads is a side-scrolling beat 'em up video game originally developed by Rare and then remastered by Dlala Studios. The story follows the exploits of the Battletoad Rash, a toad who teams up with his two brothers that are amphibious creatures, Zitz and Pimple, to defeat the evil Dark Queen and her army of minions. The game features many mechanics such as punching, kicking, and using various combos that the player will have to utilize to defeat the enemies. Problem: Battletoads provides interesting gameplay and mechanics that forces the player to make rapid iterations of their strategy throughout the game. The game has many obstacles throughout the first Act. The game does not provide clear instructions on the initial mechanics that are introduced in the first Act for the player. For instance, mechanics like invincibility, charge attacks, and healing are not taught from the start of the game but can be utilized if the player already has prior knowledge of the mechanic. Summary of tests To summarize our initial testing plan, ten participants played 30-minute sessions of Battletoads. Players were exposed to consistent variables regarding the way they play. This extends to the controller they use (Xbox controller), the difficulty of the game (Toad Difficulty), their setting (IE: couch and TV setup), audio source (Gaming headset), and finally, the participants were instructed to play the game in single-player. To validate these decisions, the players were introduced to a pre-question survey regarding the participants' familiarity with the game and then post-questions regarding the game session they conducted. Lastly, the data was recorded for further data analysis of the playtests so they can be documented for validity. Summary of results The results from all ten participants come to conclude many of the hypotheses the team previously predicted. The results of hypothesis one show that 80% of participants had a general understanding of the rules of Battletoads, which was able to understand what the story was trying to portray. The results of the second hypothesis share a direct correlation between the score and the combos which resulted in only three participants being able to achieve scores over 300,000 points. Hypothesis three’s results examine how often players utilize the ability to change characters. It was discovered that 80% of our participants were able to utilize the “Tag-in” mechanic - switching between each toad willingly. Lastly, the results of the fourth hypothesis predicted that 30% of the participants were able to understand the initial collectible mechanic but only 20% of them were able to acquire every collectible during playtesting. Summary of Inferences Inferences can be made based on the results of the ten participants. Hypothesis one was directed more towards the initial learning curve of the game in Battletoads which did not clarify many of the game's mechanics very well. From our findings, the participants were able to understand the fundamental basic mechanics of the game. For example, simple movements and attacks were less inclined to use more advanced mechanics, but that was not properly explained in the initial stages of the game. The second hypothesis aimed to better understand the combo rating system as it relates to new players in the game. The results, from the findings, show that players who could maintain a high combo count were able to acquire higher overall scores by the end of the first Act. Hypothesis three focuses more on the ability to switch your character at will, this can be used strategically to get through the levels as each toad has additional mechanics they can utilize. The results show that the participants preferred going for the character's overall appearance rather than the mechanics of the characters they brought to the game. Finally, the last hypothesis analyzed the collectible system in that players would not understand the entire system, but only would understand bits and pieces of the information left by the developers. The results of this playtest concluded that initial collectibles were picked up but later, hidden collectibles would be overlooked entirely. The reason for this is that the mechanic was not properly explained to the players at the beginning of playtesting. Introduction: In this report on the testing of usability in mechanics for the popular game Battletoads, our team aims to analyze the gameplay and usability of mechanics with ten participants that playtest the game chosen Battletoads. Our team questioned many of the initial design choices of the game in

terms of the general progression of learning new mechanics and following the linear gameplay. Battletoads provides access to most of the mechanics in the game from the start but does not explain to the player how to use them properly, this led our team to question how many initial mechanics in the game are missed due to no proper explanation. Game variables like collectibles and invincibility are not required for the game's competition, however, they are offered at the beginning of the game so the player can have an easier time completing the mission’s objectives.

Game Testing Description Background Discussion: Playtesting is a useful tool in game design used to test mechanics, theories, and hypotheses created for games. Formal individual testing is used to focus on individual gameplay when testing mechanics. Usability testing in game design is used to test how well interactions with the participants work. This method is different from playtesting because the instigators of the study have clear mechanics focused on related to the study. The data that will be collected as a group will be collected from our second circle: Trusted Friends and Family. Observations and measurements were collected in Cross-Sectional studies to conduct a study of a single instance with a single wave. Empirically speaking, performing a study of social research that is based on perception through observation makes it interesting for playtesting. The reason for this is the conducted validity of the testing that could come across a threat such as a hypothesis guessing when the playtester's behavior is predictable. Each participant will be required to play test the game in single-player mode in order for the data to reflect properly with every single group. Our participants range from ages 8 to 40 years old. There is a possibility the maturity level of each participant could impact the data. Description and Research Questions: Battletoads is a classic Beat’em up platformer that was remastered in 4K by developer Dlala Studios and published by Xbox Game Studios in 2020 for Xbox One and Microsoft Windows. The game features three toads - Rash, Zitz, and Pimple - as they travel across dimensions to rescue their fellow toads from the clutches of a mad scientist, known as Dr. Pidgel. Players can select from one of three toads at the beginning of each level and must use their unique abilities to progress through the game. Rash is the fastest and relies on long-distance running to evade enemies; Pimple is the strongest, but can’t jump as high, and Zitz can wall-jump to avoid enemy attacks; each toad also has unique abilities and characteristics. Due to the learning curve in the early stages of the game, the first research question is to find out if the instructions provided were clear and precise. Some of the mechanics available that have a major impact on gameplay are Invincibility, Health Regeneration, and Charge Attack. The second research question is to find out if the participants took advantage of each mechanic. The purpose of the third research question is to find out if the player was able to execute each mechanic. One of the most useful tools in the game is the ability to evade the attack of the enemy. The final question in relation to the first hypothesis is to find out if the Evade was useful during the Act 1 battle. All researchers believed the players will not understand the combo/rating system causing them to miss out on the collectibles in the environment. The first research question will be to find out the highest combo achieved by each player. Following that would be the purpose of the second research question is to see if the player was able to utilize the jump attack. Next, the third research question is to find out if the player noticed any of the collectibles within the environment. Then, the last question in relation to hypothesis two is to find out what was the player's score after the first Act. In the third hypothesis, it is believed that the players will not know which character is best for certain situations resulting in the player not utilizing the drop-out mechanic which allows switching between toads. Finding out if the player “tag-in” option is useful for the first research question. Which would lead to the second research question to find out which character was used the most. Afterward, the third research question is to find out which character was selected at the start of the game. Lastly, the final research question related to hypothesis three is to find out which character was favored the least and why. Battletoad does not explain the collectible system to the players very clearly as a prediction. This made the researchers believe players will miss the initial collectibles. However, the first research question would be to find out if the participants acquired all collectibles. Following this, the next research question is to see if the instructions on how to collect the collectibles were easy to understand. Finally, the last research question conducted for hypothesis four is to see if the participant was able to utilize the tongue-grab mechanic to grab collectibles. Theoretical Variables: With theoretical variables, there are many different variables that can be defined, each participant may have variables that the instigators do not understand. Although many of the variables of this test can be controlled, theoretical variables may exist in our data. The general mood along with the energy level of the participants is a factor in these tests. The ages of our participants represent a

theoretical variable that factors into the results of the data. Other theoretical variables can be broken down into a high quantity of detail which can produce various data for game testing. Operational Variables: Operation variables in our playtests need to be consistent throughout the tests. Some operational variables include the use of a TV for the participant's tests instead of a gaming monitor which participants may be more familiar with. More operational variables exist in the setting as participants will be featured on a couch with a gaming headset on. These variables are more easily controlled but still can skew the results based on theoretical variables present in the participants. Hypotheses: The goal of these playtest sessions was to validate the initial hypotheses the team formulated regarding the usability of mechanics offered to the player. The learning curve of the game was hypothesized to be difficult and did not provide enough details in terms of mechanical usefulness. When it comes to the first hypothesis, our team will focus on four specific areas: Are the instructions clear and precise, was the player able to take advantage of each mechanic, did the player execute each mechanic, and was the evade mechanic useful during Act 1 Scene 1 battle. Secondly, it was hypothesized that players would not understand the combo/rating system which is a factor in the collection of all collectibles in each stage. Questions regarding the combo mechanic will be used to further this hypothesis providing insight into its validity based on the results. Next, our team questioned if players could differentiate between the different character abilities in picking the best viable option in terms of character selection for each stage. To test this, our team will gather user feedback as well as recorded gameplay that will in-depth analyze the rationale behind character decisions in Battletoads. Lastly, it was hypothesized that the collectibles in the game were not introduced or locked behind unexplained mechanics causing them to be missed by the players. Through observation, our team will be able to observe which collectibles are missed and collected while still holding the integrity of validity in that players will be playing blindly through the game with no overarching goals designated by the test team. All of these hypotheses are in regard to the usability of the game and its design in the progression of mechanics. The variables tested in these playtests will provide insight into player choice based on their own experience of the game.

Analysis methods To test the hypothesis our team created, both a Pre-test and a Post-test were administered to the players before and after playtesting. Utilizing “Google Forms”, our survey questions are laid out in multiple choice and short answer forms. The goal is to gather the results from each question of the pre-test and post-test surveys. Pre-Test: Players were given the survey to complete which consisted of general questions relating to the participant's past gaming experience. Each participant is first asked about their experience with the game Battletoads so we can get a better understanding of their experience with the game itself. The participants are then asked about their experience with the side-scroller genre. Next, players are asked several questions pertaining to how they game. What is their preferred gaming device? How many hours do the participants game a week? What sitting position do they prefer? And finally, what time of the day do the participants enjoy playing games? These questions are posed to get a better understanding for each participant's experience in relation to new mechanics they may experience in the game Battletoads.
Post-Test: After our participants experienced the playtest our team set up, they were administered Post-test questions, a similar format to the pre-test, that relate to the gameplay they just played. The questions were divided into four sections that represented each hypothesis the team developed. Within the division of each section, questions are broken down into detailed questions that are either multiple choice or short answers. The first hypothesis was geared towards the instructions and learning of mechanics in Battletoads. Players were asked if the instructions were clear and concise so we could get a better understanding where the problem with mechanics arose. Next, the participants were given a list of mechanics available in the game. They were then asked if they used all the mechanics in their own playtest. Afterward, the participants were then asked if they were able to use the “Evade” mechanic. Lastly, the participants were asked if they utilized all available mechanics in the game during their playtests. The second hypothesis asked the participants if they utilized the jump attack which led to combos in the game. Next, participants were asked about their highest combo/ highest score achieved in their playthrough so the team could get a good idea if the combo system related to the collections system. Finally, the participants are asked about the hidden collectibles located within the game. With that said, supporting the third hypothesis, players were asked questions regarding the options of each toad in the game. Furthermore, the participants were asked which toad was their favorite and least favorite to gain an understanding of why they were initially chosen. Next, participants are asked if they utilized the “Tag-in” mechanic in their playtests to test which toads were used most often.

The final section of questions relating to the fourth hypothesis asks if the collection system was explained properly. Participants are then asked if the instructions regarding collectables were clear during their playtest. Next, the participants are asked if they were able to acquire all of the collectibles during their tests. Which would lead to the last question if they were able to utilize the tongue-grab mechanic to ensure they collected the hidden collectibles.

Pre-Test Results The results for the pretest, where the pre-test questions follow, was to find out more background information about each participant. These results are listed in the form of graphs and pie charts. Providing these pretest questions to our participants proves how diverse our participants are based on their ages ranging between 8 to 40 years old. With that said, the inference of the pre-test would share the results broken down into detail. Post-Test Results Hypothesis 1 Results: Hypothesis One stated that Battletoads has a learning curve at the beginning of the game due to lack of instruction and causes players to not use all of their available mechanics. Results show that 80% claim that the instructions were clear, but still 70% of participants claim to not use all available mechanics. 80% of the players reported that they were able to utilize the evade mechanic. Hypothesis 2 Results: Hypothesis Two stated players will not understand the combo/rating system causing them to miss the collectibles. Jump attack appears to be an easy mechanic to grasp with 90% of players reporting they were able to utilize the mechanic. With only 20% of the participants able to achieve a combo streak over 50, results show that the participant did not clearly understand the system. Results show 60% of the participants did not notice the collectibles during battle in the environment. Hypothesis 3 Results: Hypothesis 3 identifies problems with the character selection as all 3 toads come with different abilities unique to their design. 40% of participants used the toad Rash whereas only 30% of participants used their counterpart toad options. Rash was also the most initially picked at 66% whereas Zits and Pimple were used at 22% and 11% respectively. 80% of players reported having used the Tag-in mechanic to switch their characters during the game. Finally, Pimple was the least favorite character with 60% of participants not enjoying his sluggish design and slow move set. 30% did not enjoy Zits and only a mere 10% disliked the popular toad Rash. Hypothesis 4 Results: Finally, hypothesis 4 explores the collectible systems. Of the 10 participants who playtested Battletoads, 30% of players stated that they understood the collectible system as first initially explained. 20% had somewhat of an understanding of the system but the last 50% of players were left not fully understanding the collectible system. Only 20% of the participants were able to collect all of the available collectibles and only 50% of the participants utilized the tongue-grab mechanic in the game used to collect. Pre-Test Inferences: Inferences can be made from the results of the participant's pre-test. With 80% of the participants never having played Battletoads before, it is likely that many of the players would not understand the more advanced mechanics of the game. 60% of players had experience with side scrollers before which helped them navigate the map using many of the general mechanics in side scroller games. Finally, most of our participants played on devices that differ from the Xbox so mechanical controls would be new to some of the players. Post-Test Inferences: Hypothesis 1 Inferences: Although 80% of players reported a clear understanding of the instructions, 70% of all participants were not able to take advantage of each combat mechanic (Evade, Invincibility, Health Regeneration, Charge Attack). The basic mechanics of the game were understood but the combat-based mechanics are not explained thoroughly to the player causing them to not be able to utilize all available skills in combat. The evade mechanic turns out to be useful with 80% of the participants utilizing the evade mechanic in the first act. This infers that the evade mechanic was properly explained with sidenote tutorials, so most of the participants were able to utilize this mechanic throughout the level. Finally, All of the participants reported that they did not utilize every available mechanic at their disposal after playing for 30-minute sessions. It is inferred that the more complex mechanics of the game did not receive tutorial sidenotes causing players to forget/not to utilize certain mechanics available Hypothesis 2 Inferences: The Jump attacks are crucial to obtaining high combos in Battletoads; 90% of players were able to utilize this mechanic to extend their combo scores. It is inferred that participants that utilized the jump attack received higher overall combos/scores during their playtests. Participants were tasked with locating hidden collectibles. 60% of the participants did not notice the collectibles hidden in the environment. This is because the acquisition of collectibles hidden in the level and behind combo scores are not explained clearly so the players missed them throughout the playtesting. Only 20% of players were able to achieve a combo streak over 50, causing only 3 participants to obtain a final score of over 300,000. Only 3 participants were able to unlock the combo collectible at the end of the level,

This is because the objectives to collect this collectible were not clear. Finally, 60% of the participants were not able to complete the first Act in Battletoads resulting in a lower score. One of the collectibles is dependent on the player's completion of the first act; This caused most of the participants to not be able to acquire this collectible. Hypothesis 3 Inferences: Reports show that 80% of players utilized the Tag-in mechanic during gameplay. The test results show that 40% of the participants chose Rash as the character used the most, proving the player prefers rash over the other toads. Reports show at 66%, Rash was the character most players started with initially. Players initially enjoyed the design of Rash. Participants reacted visually to the design of Rash making him the most desirable toad for playtesting the game. Reports show that 60% of the players chose Pimple as the least favorite of the characters. Players did not enjoy the size and the speed of this toad in comparison to the other two who had similar move sets. Hypothesis 4 Inferences: 30% of the players reported having a clear understanding of the collectibles whereas 50% did not have a clear understanding and 20% had somewhat of an understanding. Players understand the initial collection mechanic in that they must walk over the object to collect it. 80% of the participants reported they did not collect all the available collectibles in their play-through. Although players were able to understand the initial instruction to pick up collectibles, Battletoads does not provide a tutorial for some of the hidden collectibles causing players to miss them. 50% of the players utilized the tongue-grab mechanic. The tongue-grab mechanic was used to collect hidden collectibles in Battletoads. Players that did not utilize this mechanic to the full extent were unable to collect all of the collectibles in their playtest. Limitations Validity Limitations: The testing performed for the game will conclude an expectation for Battletoads. A few limitations will be discovered in game testing such as gender differences, age, and setting for the testing. As part of the external validity, the data would include a threat of the place for the testing as well as any varied equipment from the instigators of the test (EX: Gaming headset, TV aspects). In terms of statistical conclusion validity, There is a possibility of a false negative result if the team is unable to pinpoint the main issue in the explanation of the instructions in the game Battletoads. Conclusion Summarized Findings: To summarize the results from the Battletoads playtests, Players were set up in a closed environment to determine how well this game can be played, which toad is utilized the most, as well as to see how well each participant did by the end of the test time of 30 minutes. Each participant was set up congruently in a living room environment with a tv and Xbox controller. To better the participants' experience, a variable for the test playing was to have a headset to have the player have the ultimate experience. The players were given surveys before play testing to understand the background of each participant and then given afterward to fulfill each research question developed. Each hypothesis consisted of four to three questions, provided by “Google Forms”. The experiment consisted of ten participants that ranged from 8 to 40 years old of both genders. Suggestions for Future Research In future research, it is important to have a limited range of answers for the Pre-test and Post-test when asking questions regarding a specific mechanic. For example, when asking why a character is someone's least favorite, suggest 5 different answers go off for the participants instead of leaving it open-ended. Another suggestion for future research would be to explore other ways to play the game and see if that has any effect on the participant's performance. Finally, future research should consist of testing other conditions regarding the participants preferred playstyle in terms of equipment and seating arrangement to compare the results. Usability and Testing Usability is important to test in video game production. Test to ensure that the player base is interacting with the designed mechanics of the game properly. Generally speaking, there should be a player interaction every 8 seconds in video games so having a strong method for usability testing provides a lot of insight for developers in terms of where they should put their efforts when designing a scene. Playtesting also provides a lot of information regarding interactions that players will experience in a game. Administers of playtests are able to gather information regarding the setting, player enjoyment, player preference, and other important information used to design a video game. Testing is a highly valuable skill for production teams and applying strong and effective methods of testing can provide insight for your team that may have been overlooked in the original design.

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